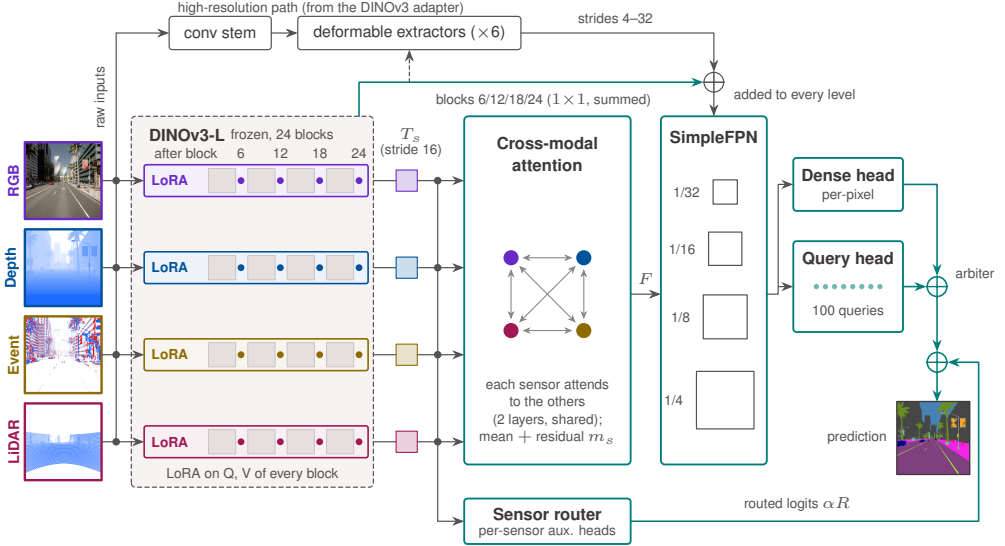

One frozen backbone, sensor-specific adapters, cross-modal fusion
high-resolution path (from the DINOv3 adapter)
conv stem
deformable extractors (×6)
strides 4–32
raw inputs
blocks 6/12/18/24 (1 × 1, summed)
added to every level
DINOv3-L frozen, 24 blocks
after block 6 12 18 24
T_s (stride 16)
RGB
Depth
Event
LiDAR
LoRA
LoRA
LoRA
LoRA
LoRA on Q, V of every block
Cross-modal attention
each sensor attends to the others (2 layers, shared); mean + residual m_s
F
SimpleFPN
1/32
1/16
1/8
1/4
Dense head
per-pixel
Query head
100 queries
arbiter
prediction
Sensor router
per-sensor aux. heads
routed logits αR